

FINAL EXPLANATION: GENERAL SCIENCE GRADE 10

Student Assessment Document

FINAL EXPLANATION: GENERAL SCIENCE GRADE 10 – SUB-STRAND 1.1: INTRODUCTION TO GENERAL SCIENCE – FINAL EXPLANATION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

This is your Final Explanation for Sub-strand 1.1. Use everything you have learned from the Kisumu maize-farmer puzzle, your research, the Mystery Bags activity, your Driving Question Board and your Cumulative Science Model. Answer every section in the blank space under each prompt. Write in full sentences. Use the words evidence, reasoning and inference correctly, and give Kenyan examples. Driving Question: How does a scientist know something is true when they cannot see it directly, and how is this power of General Science shaping life, careers, and the environment in Kenya?

Section 1: Explaining the Kisumu Farmer Phenomenon

Recall the maize farmer in Kisumu whose plants along one edge of the shamba were wilting and yellowing even though she watered and fertilised all the plants equally. She never saw any insect on the plants, yet her conclusion about the cause turned out to be correct. (a) List the pieces of evidence she used. (b) Describe the reasoning that connected her evidence to her conclusion. (c) State her inference, including what the hidden cause was and where it was. (d) Explain why her conclusion was not just a guess.

Write your answer here:

Section 2: What General Science Is and Why It Matters

(a) In your own words, say what General Science is, and explain why it is more than a collection of facts. (b) Choose TWO of the three domains: human life, environment, technology. For each, give one Kenyan example of General Science at work, and say what hidden thing the people involved had to infer or detect.

Write your answer here:

Section 3: The Principle of Inference

(a) Define inference. (b) Name and describe the three parts of the evidence-reasoning-inference (ERI) framework. (c) Choose ONE of these Kenyan scenarios: Dr. Kamau reading a chest X-ray at Kenyatta National Hospital, or Warden Lesiamon tracking a poacher in Amboseli. Complete an ERI analysis for it.

Write your answer here:

Section 4: Collecting Evidence and Process Skills

In the Mystery Bags activity you used written clue cards to work out a hidden cause on a Kenyan farm. (a) Explain the difference between an observation and an inference, with an example. (b) Name at least three science process skills your group used and say how you used each one. (c) What should a scientist do when one clue contradicts the others, and why?

Write your answer here:

Section 5: Careers, a New Scenario and Your Growth

A tea farmer in Kericho notices that the leaves on some of her tea bushes are curling. She waters and cares for all her bushes in the same way. The curling leaves are mostly on the bushes in one part of the field, the other bushes look healthy, and the curling began during the dry season. She cannot see any pest or disease on the leaves. (a) Identify two pieces of evidence from this scenario. (b) State one reasonable inference about what might be causing the curling, and explain your reasoning from the evidence. (c) Name a General Science career in Kenya that would investigate this professionally and say how that person would use inference. (d) Finally, answer the Driving Question in 3 to 4 sentences, explaining how your thinking has changed since Lesson 1.

Write your answer here:

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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Use of evidence from the lessons	Uses specific evidence from the Kisumu farmer, the Kenyan examples and the Mystery Bags activity accurately and links each to the answer.	Uses some evidence from the lessons correctly, but links to the answer are sometimes unclear.	Gives few or vague examples, or uses evidence that is incorrect or not from the lessons.
Understanding of inference	Defines inference correctly, explains evidence, reasoning and inference clearly, and explains why inference is not a guess.	Defines inference and names the three parts, with limited explanation of how they connect.	Confuses inference with guessing or observation, or cannot name the parts.
Application to a new scenario	Applies the ERI framework to the Kericho tea scenario using the evidence given, a logical inference about a hidden cause that follows from that evidence, and a	Applies the framework to the new scenario but the reasoning or career link is incomplete.	Gives an inference without evidence or reasoning, or does not name a career.

	suitable career.		
Science process skills and Kenyan contexts	Correctly names and explains at least three process skills and the difference between observation and inference, with relevant Kenyan contexts throughout.	Names two or three process skills with some explanation and includes some Kenyan examples.	Names few or no process skills, or examples are missing or not relevant.
Answer to the Driving Question and communication	Gives a clear, complete answer to the Driving Question that shows how thinking has changed, written in clear, organised sentences.	Answers the Driving Question partly, with some reflection on change in thinking.	Answer to the Driving Question is incomplete or unclear, with no reflection.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

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